

TPG4190 Seismic data acquisition and processing Imaging

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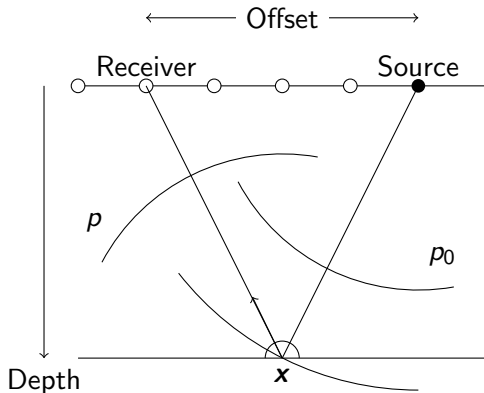
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Trondheim fall 2023

Overview

- ▶ Imaging conditions
- ▶ Reciprocity
- ▶ Time reversal
- ▶ Imaging formula

Imaging condition I

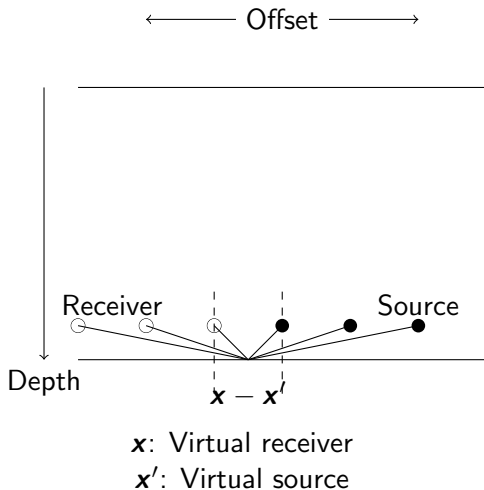


p : Scattered wavefield (data)

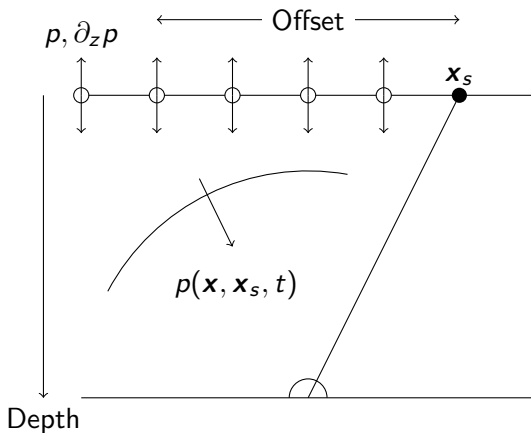
p_0 : Modeled wavefield

x : Spatial position

Imaging condition II



Imaging condition III

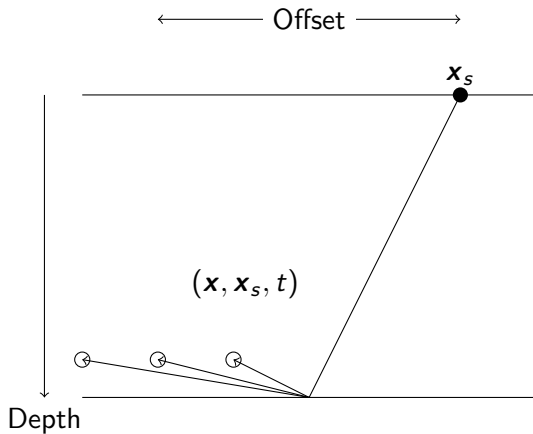


p : Scattered wavefield

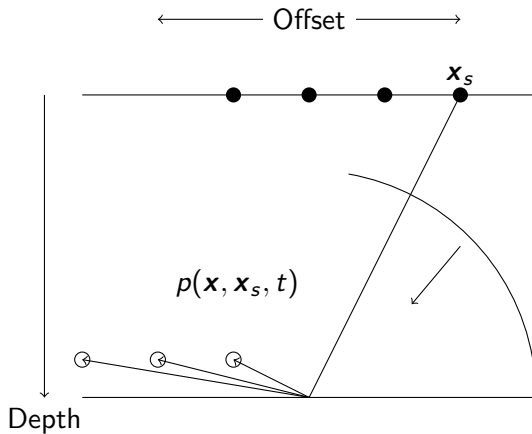
$\mathbf{x}_s$: Source position

$\mathbf{x}, t$: Position, time

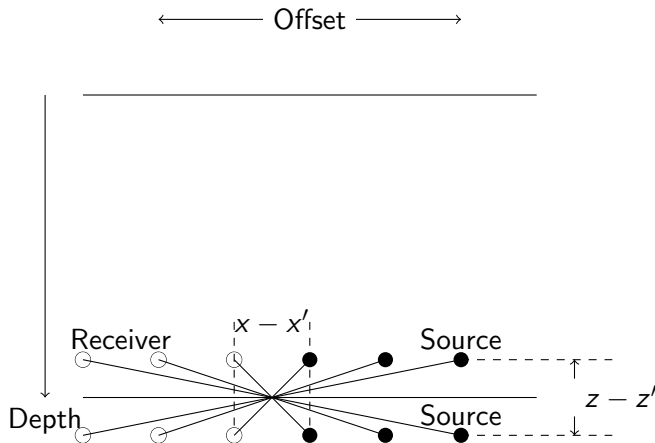
Imaging condition IV



Imaging condition V



Imaging condition VI



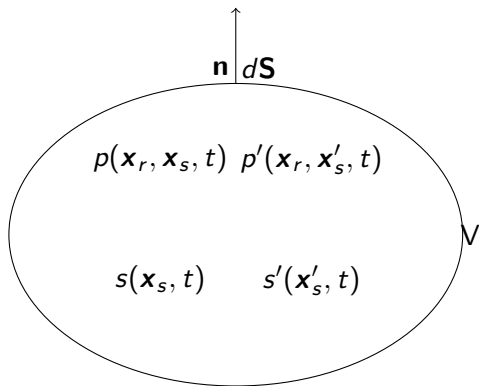
$x - x'$: Horizontal offset

$z - z'$: Vertical offset

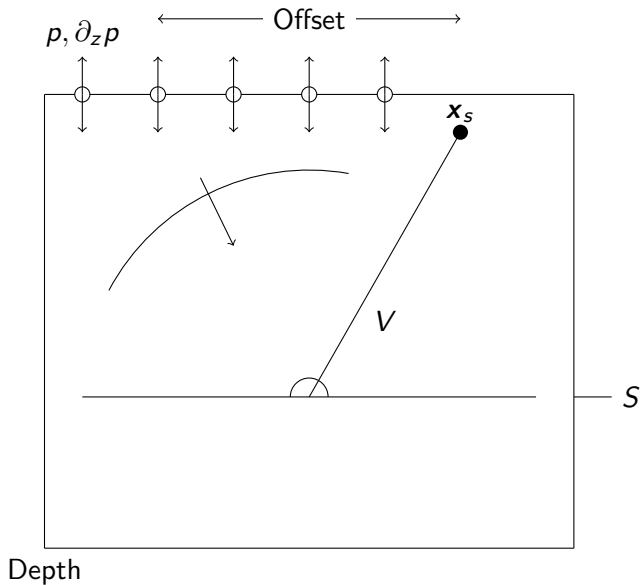
Reciprocity

Forward modeling

Reciprocity



Reciprocity



Reciprocity

Two sources s' and s are located inside a volume V with surface S at positions $\mathbf{x}_s$ and $\mathbf{x}'_s$. The corresponding wavefields are $p(\mathbf{x}, \mathbf{x}_s)$ and $p'(\mathbf{x}, \mathbf{x}'_s)$. The wave equations for these two fields are

$$\nabla^2 p(\mathbf{x}, \mathbf{x}_s, t) - \frac{1}{c^2(\mathbf{x})} \partial_t^2 p(\mathbf{x}, \mathbf{x}_s, t) = s(\mathbf{x}, t), \quad (1)$$

$$\nabla^2 p'(\mathbf{x}, \mathbf{x}'_s, t) - \frac{1}{c'^2(\mathbf{x})} \partial_t^2 p'(\mathbf{x}, \mathbf{x}'_s, t) = s'(\mathbf{x}, t) \quad (2)$$

Reciprocity

Fourier transform over time ($\partial_t^2 \rightarrow -\omega^2$)

$$\nabla^2 P(\mathbf{x}, \mathbf{x}_s, \omega) + \frac{\omega^2}{c^2(\mathbf{x})} P(\mathbf{x}, \mathbf{x}_s, \omega) = S(\mathbf{x}, \omega), \quad (3)$$

$$\nabla^2 P'(\mathbf{x}, \mathbf{x}'_s, \omega) + \frac{\omega^2}{c'^2(\mathbf{x})} P'(\mathbf{x}, \mathbf{x}'_s, \omega) = S'(\mathbf{x}, \omega) \quad (4)$$

Reciprocity

Multiply equation (3) with P' and equation (4) with P and integrate over V (suppressing ω as an argument)

$$\int dV(\mathbf{x}) \left[P'(\mathbf{x}, \mathbf{x}'_s) \nabla^2 P(\mathbf{x}, \mathbf{x}_s) + P'(\mathbf{x}, \mathbf{x}'_s) \frac{\omega^2}{c^2(\mathbf{x})} P(\mathbf{x}, \mathbf{x}_s) \right] = \int dV(\mathbf{x}) P'(\mathbf{x}, \mathbf{x}'_s) S(\mathbf{x}) \quad (5)$$

$$\int dV(\mathbf{x}) \left[P(\mathbf{x}, \mathbf{x}_s) \nabla^2 P'(\mathbf{x}, \mathbf{x}'_s) + P(\mathbf{x}, \mathbf{x}_s) \frac{\omega^2}{c'^2(\mathbf{x})} P'(\mathbf{x}, \mathbf{x}'_s) \right] = \int dV(\mathbf{x}) P(\mathbf{x}, \mathbf{x}_s) S'(\mathbf{x}) \quad (6)$$

Reciprocity

Subtract equation (6) from equation (5): (Assume $c' = c$).

$$\begin{aligned} \int dV(\mathbf{x}) \left[P'(\mathbf{x}, \mathbf{x}'_s) \nabla^2 P(\mathbf{x}, \mathbf{x}_s) - P(\mathbf{x}, \mathbf{x}_s) \nabla^2 P'(\mathbf{x}, \mathbf{x}'_s) \right] = \\ \int dV(\mathbf{x}) \left[P'(\mathbf{x}, \mathbf{x}'_s) S(\mathbf{x}) - P(\mathbf{x}, \mathbf{x}_s) S'(\mathbf{x}) \right] \quad (7) \end{aligned}$$

Reciprocity

Gauss divergence theorem:

$$\int dV(\mathbf{x}) \nabla \cdot \mathbf{A}(\mathbf{x}) = \int dS \mathbf{A}(\mathbf{x}) \cdot \mathbf{n}(\mathbf{x}) \quad (8)$$

Put $\mathbf{A} = P'(\mathbf{x}, \mathbf{x}'_s) \nabla P(\mathbf{x}, \mathbf{x}_s)$ to get

$$\begin{aligned} \int dV(\mathbf{x}) \nabla \cdot [P'(\mathbf{x}, \mathbf{x}'_s) \nabla P(\mathbf{x}, \mathbf{x}_s)] &= \\ \int dV(\mathbf{x}) [P'(\mathbf{x}, \mathbf{x}'_s) \nabla^2 P(\mathbf{x}, \mathbf{x}_s) + \nabla P'(\mathbf{x}, \mathbf{x}'_s) \cdot \nabla P(\mathbf{x}, \mathbf{x}_s)] &= \\ \int dS P'(\mathbf{x}, \mathbf{x}'_s) \nabla P(\mathbf{x}, \mathbf{x}_s) \cdot \mathbf{n}(\mathbf{x}) & \quad (9) \end{aligned}$$

Reciprocity

Put $\mathbf{A} = P(\mathbf{x}, \mathbf{x}_s) \nabla P'(\mathbf{x}, \mathbf{x}'_s)$ to get

$$\begin{aligned} \int dV(\mathbf{x}) \nabla \cdot [P(\mathbf{x}, \mathbf{x}_s) \nabla P'(\mathbf{x}, \mathbf{x}'_s)] &= \\ \int dV(\mathbf{x}) [P(\mathbf{x}, \mathbf{x}_s) \nabla^2 P'(\mathbf{x}, \mathbf{x}'_s) + \nabla P'(\mathbf{x}, \mathbf{x}'_s) \cdot \nabla P(\mathbf{x}, \mathbf{x}_s)] &= \\ \int dS P(\mathbf{x}, \mathbf{x}_s) \nabla P'(\mathbf{x}, \mathbf{x}'_s) \cdot \mathbf{n}(\mathbf{x}) & \quad (10) \end{aligned}$$

Subtract equation (10) from equation (9) to get

$$\begin{aligned} \int dV(\mathbf{x}) [P'(\mathbf{x}, \mathbf{x}'_s) \nabla^2 P(\mathbf{x}, \mathbf{x}_s) - P(\mathbf{x}, \mathbf{x}_s) \nabla^2 P'(\mathbf{x}, \mathbf{x}'_s)] &= \\ \int dS [P'(\mathbf{x}, \mathbf{x}'_s) \nabla P(\mathbf{x}, \mathbf{x}_s) - P(\mathbf{x}, \mathbf{x}_s) \nabla P'(\mathbf{x}, \mathbf{x}'_s)] \cdot \mathbf{n}(\mathbf{x}) & \quad (11) \end{aligned}$$

Reciprocity

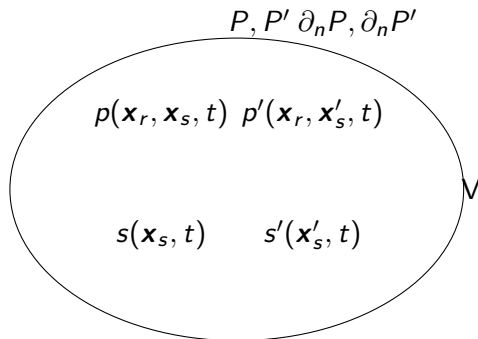
Replace the left hand side of equation (7) with equation (11) to get

$$\int dS(\mathbf{x}) [P'(\mathbf{x}, \mathbf{x}'_s) \nabla P(\mathbf{x}, \mathbf{x}_s) - P(\mathbf{x}, \mathbf{x}_s) \nabla P'(\mathbf{x}, \mathbf{x}'_s)] \cdot \mathbf{n}(\mathbf{x}) = \int dV(\mathbf{x}) [P'(\mathbf{x}, \mathbf{x}'_s) S(\mathbf{x}) - P(\mathbf{x}, \mathbf{x}_s) S'(\mathbf{x})] \quad (12)$$

Assume point source $S'(\mathbf{x}) = \delta(\mathbf{x} - \mathbf{x}'_s)$ to get

$$P(\mathbf{x}'_s, \mathbf{x}_s) = \int dV(\mathbf{x}) P'(\mathbf{x}, \mathbf{x}'_s) S(\mathbf{x}) - \int dS(\mathbf{x}) [P'(\mathbf{x}, \mathbf{x}'_s) \nabla P(\mathbf{x}, \mathbf{x}_s) - P(\mathbf{x}, \mathbf{x}_s) \nabla P'(\mathbf{x}, \mathbf{x}'_s)] \cdot \mathbf{n}(\mathbf{x}) \quad (13)$$

Reciprocity



Reciprocity

$P, P' \quad \partial_n P, \partial_n P'$

$p(\mathbf{x}_r, \mathbf{x}_s, t) \quad p'(\mathbf{x}_r, \mathbf{x}'_s, t)$

$s(\mathbf{x}_s, t) \quad s'(\mathbf{x}'_s, t)$

V

Source-receiver reciprocity

Assume now that the surface integral is zero and that $S(\mathbf{x}) = \delta(\mathbf{x} - \mathbf{x}_s)$. Equation (12) then gives if we assume $P = P'$ and $S = S'$

$$P(\mathbf{x}_s, \mathbf{x}'_s) = P(\mathbf{x}'_s, \mathbf{x}_s) \quad (14)$$

The pressure recorded at $\mathbf{x}_s$ due to a source at $\mathbf{x}'_s$ is the same as the pressure recorded at $\mathbf{x}'_s$ due to a source at $\mathbf{x}_s$.

Reciprocity

When the source is a delta function

$$S(\mathbf{x}) = \delta(\mathbf{x} - \mathbf{x}_s), \quad (15)$$

the pressure P is usually called a Green's function

$$P(\mathbf{x}, \mathbf{x}_s) = G(\mathbf{x}, \mathbf{x}'_s)$$

Reciprocity

Use reciprocity $G(\mathbf{x}'_s, \mathbf{x}) = G(\mathbf{x}, \mathbf{x}'_s)$ to get from equation (13)

$$P(\mathbf{x}'_s, \mathbf{x}_s) = \int dV(\mathbf{x}) G(\mathbf{x}'_s, \mathbf{x}) S(\mathbf{x}) \\ - \int dS(\mathbf{x}) [G(\mathbf{x}'_s, \mathbf{x}) \nabla P(\mathbf{x}, \mathbf{x}_s) - P(\mathbf{x}, \mathbf{x}_s) \nabla G(\mathbf{x}'_s, \mathbf{x})] \cdot \mathbf{n}(\mathbf{x})$$

Rename $\mathbf{x} \rightarrow \mathbf{x}'$ and $\mathbf{x}'_s \rightarrow \mathbf{x}$.

$$P(\mathbf{x}, \mathbf{x}_s) = \int dV(\mathbf{x}') G(\mathbf{x}, \mathbf{x}') S(\mathbf{x}') \\ - \int dS(\mathbf{x}') [G(\mathbf{x}, \mathbf{x}') \nabla P(\mathbf{x}', \mathbf{x}_s) - P(\mathbf{x}', \mathbf{x}_s) \nabla G(\mathbf{x}, \mathbf{x}')] \cdot \mathbf{n}(\mathbf{x}')$$

Reciprocity

Fourier transform back to time

$$\begin{aligned} p(\mathbf{x}, \mathbf{x}_s, t) = & \int dV(\mathbf{x}') g(\mathbf{x}, \mathbf{x}', t) * s(\mathbf{x}', t) \\ & - \int dS(\mathbf{x}') \left[g(\mathbf{x}, \mathbf{x}', t) * \nabla p(\mathbf{x}', \mathbf{x}_s, t) \right. \\ & \left. p(\mathbf{x}', \mathbf{x}_s, t) * \nabla g(\mathbf{x}, \mathbf{x}', t) \right] \cdot \mathbf{n}(\mathbf{x}') \end{aligned}$$

Forward modeling

If the surface integral can be neglected

$$p(\mathbf{x}, \mathbf{x}_s, t) = \int dV(\mathbf{x}') g(\mathbf{x}, \mathbf{x}', t) * s(\mathbf{x}', t) \quad (16)$$

Reciprocity

The interpretation of this equations is

1. The volume integral describes the contribution to the pressure from the source $S(\mathbf{x})$ inside the volume V .
2. The surface integral describes the effect of the boundary conditions at the boundary S .

Time reversal

Two sources s' and s' are located inside a volume V with surface S at positions $\mathbf{x}_s$ and $\mathbf{x}'_s$. The corresponding wavefields are $p(\mathbf{x}, \mathbf{x}_s, t)$ and $p'(\mathbf{x}, \mathbf{x}'_s, -t)$. The wave equations for these two fields are

$$\begin{aligned}\nabla^2 p(\mathbf{x}, \mathbf{x}_s, t) - \frac{1}{c^2(\mathbf{x})} \partial_t^2 p(\mathbf{x}, \mathbf{x}_s, t) &= s(\mathbf{x}, t), \\ \nabla^2 p'(\mathbf{x}, \mathbf{x}'_s, -t) - \frac{1}{c^2(\mathbf{x})} \partial_t^2 p'(\mathbf{x}, \mathbf{x}'_s, -t) &= s'(\mathbf{x}, -t),\end{aligned}\tag{17}$$

Time reversal

Assume point sources $S'^*(\mathbf{x}) = H'^*\delta(\mathbf{x} - \mathbf{x}'_s)$ and $S(\mathbf{x}) = H\delta(\mathbf{x} - \mathbf{x}'_s)$ in equation (12)

$$\int dS(\mathbf{x}) [P'^*(\mathbf{x}, \mathbf{x}'_s)\nabla P(\mathbf{x}, \mathbf{x}_s) - P(\mathbf{x}, \mathbf{x}_s)\nabla P'^*(\mathbf{x}, \mathbf{x}'_s)] \cdot \mathbf{n}(\mathbf{x}) = P'^*(\mathbf{x}_s, \mathbf{x}'_s)H - P(\mathbf{x}'_s, \mathbf{x}_s)H'^* \quad (18)$$

Use reciprocity to get:

$$P'^*(\mathbf{x}_s, \mathbf{x}'_s)H - P(\mathbf{x}_s, \mathbf{x}'_s)H'^* = \int dS(\mathbf{x}) [P'^*(\mathbf{x}, \mathbf{x}'_s)\nabla P(\mathbf{x}_s, \mathbf{x}) - P(\mathbf{x}_s, \mathbf{x})\nabla P'^*(\mathbf{x}, \mathbf{x}'_s)] \cdot \mathbf{n}(\mathbf{x}) \quad (19)$$

Imaging

Renaming $\mathbf{x}_s \rightarrow \mathbf{x}$, $\mathbf{x} \rightarrow \mathbf{x}'$ and $\mathbf{x}'_s \rightarrow \mathbf{x}_s$

$$\begin{aligned} P'^*(\mathbf{x}, \mathbf{x}_s)H - P(\mathbf{x}, \mathbf{x}_s)H'^* = \\ \int dS(\mathbf{x}') \left[P'^*(\mathbf{x}', \mathbf{x}_s) \nabla P(\mathbf{x}, \mathbf{x}') - P(\mathbf{x}, \mathbf{x}') \nabla P'^*(\mathbf{x}', \mathbf{x}_s) \right] \cdot \mathbf{n}(\mathbf{x}') \end{aligned} \quad (20)$$

Fourier transforming back to time gives:

$$\begin{aligned} p(\mathbf{x}, \mathbf{x}_s, -t) * h(t) - p(\mathbf{x}, \mathbf{x}_s, t) * h'(-t) = \\ \int dS(\mathbf{x}') \left[p(\mathbf{x}', \mathbf{x}_s, -t) * \nabla p(\mathbf{x}, \mathbf{x}', t) \right. \\ \left. - p(\mathbf{x}, \mathbf{x}', t) * \nabla p(\mathbf{x}', \mathbf{x}_s, -t) \right] \cdot \mathbf{n}(\mathbf{x}') \end{aligned} \quad (21)$$

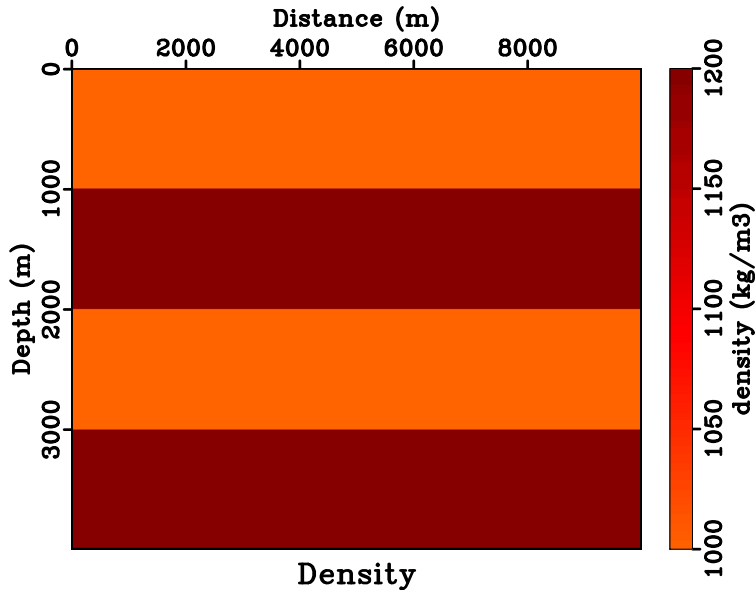
In the last equation I have dropped the prime on p , since the time-reversed field is identified by the negative time argument.

Imaging

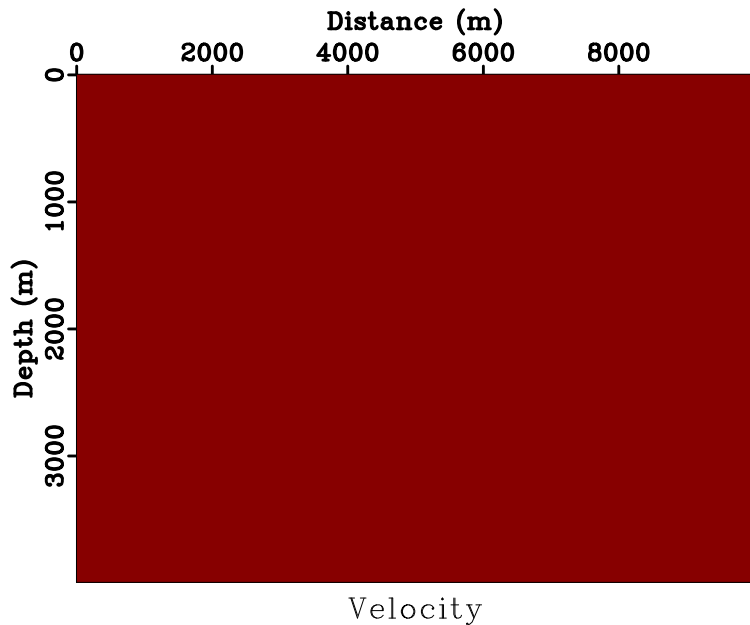
Interpretation:

1. The lefthand side of equation (21) is the pressure p for time $t > 0$ and time $t < 0$. The pressure for $t > 0$ is zero for $t < 0$ and the pressure for $t < 0$ is zero for $t > 0$. Both pressures are obtained at a field point $\mathbf{x}$.
2. The pressure for negative t on the right hand side describes time-reversed data due to a source at $\mathbf{x}_s$ recorded at the surface S . The pressure for positive time on the right hand side describes modeling of data from the receiver positions to an arbitrary field point $\mathbf{x}$.
3. The effect of the convolutions and the surface integral is to extrapolate the recorded data at the surface S to an arbitrary field point $\mathbf{x}$.

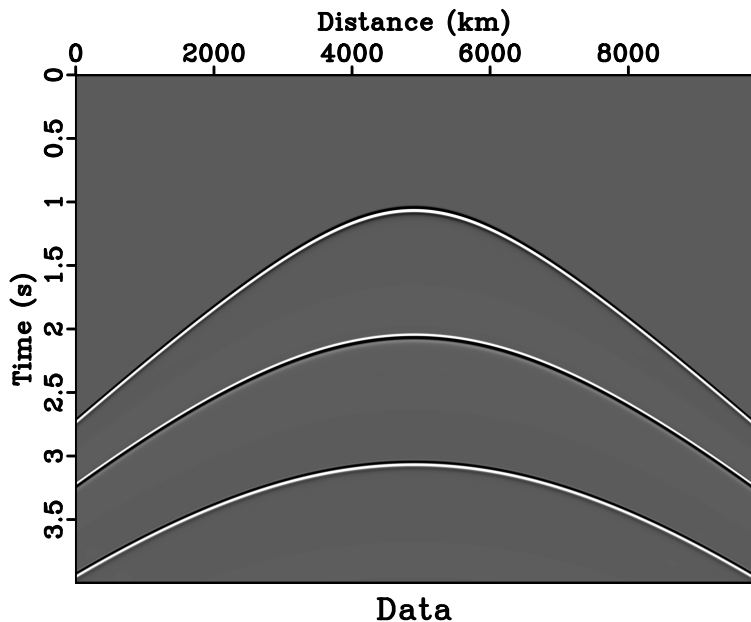
Numerical example



Numerical example

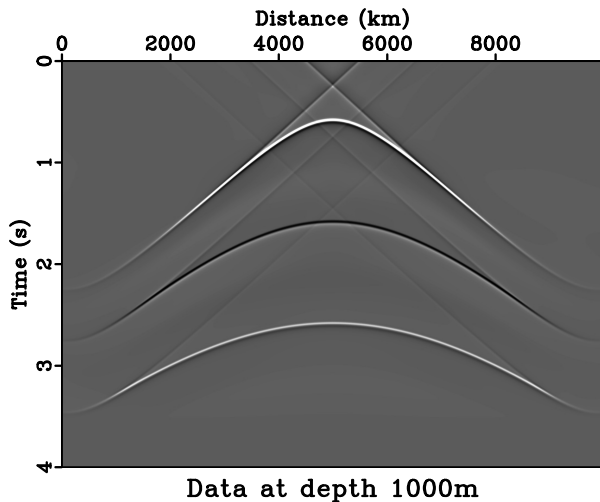


Numerical example

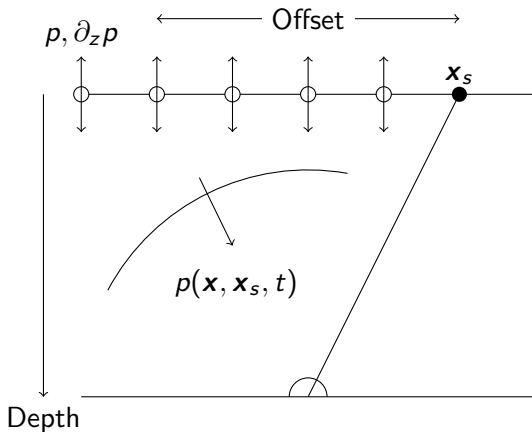


Numerical example

$p(\mathbf{x}, \mathbf{x}_s, t)$ at depth of 1000 m.



Imaging condition III

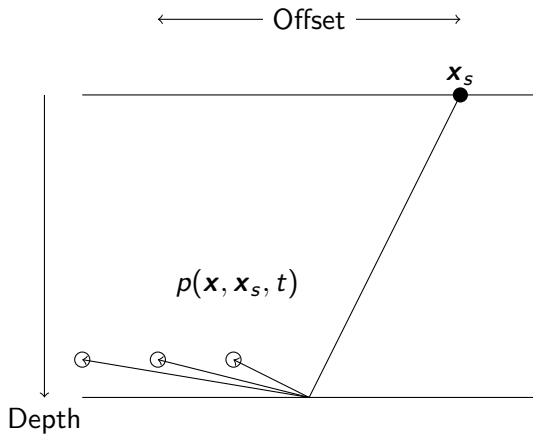


p : Scattered wavefield

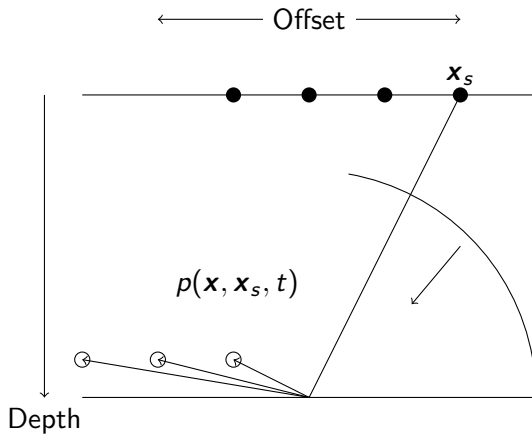
$\mathbf{x}_s$: Source position

$\mathbf{x}, t$: Position, time

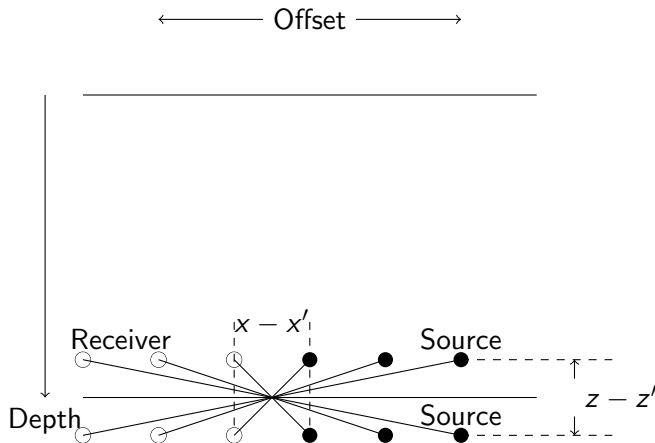
Imaging condition IV



Imaging condition V



Imaging condition VI



$x - x'$: Horizontal offset

$z - z'$: Vertical offset

Imaging

$$\begin{aligned} p(\mathbf{x}, \mathbf{x}_s, -t) * h(t) - p(\mathbf{x}, \mathbf{x}_s, t) * h(-t) = \\ \int dS(\mathbf{x}') \left[p(\mathbf{x}', \mathbf{x}_s, -t) * \nabla p(\mathbf{x}, \mathbf{x}', t) \right. \\ \left. - p(\mathbf{x}, \mathbf{x}', t) * \nabla p(\mathbf{x}', \mathbf{x}_s, -t) \right] \cdot \mathbf{n}(\mathbf{x}') \end{aligned} \quad (22)$$

Use reciprocity again

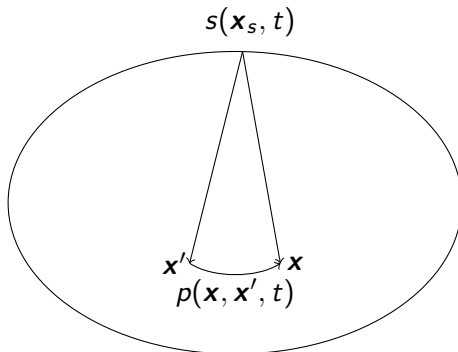
$$\begin{aligned} p(\mathbf{x}_s, \mathbf{x}, -t) * h(t) - p(\mathbf{x}_s, \mathbf{x}, t) * h(-t) = \\ \int dS(\mathbf{x}') \left[p(\mathbf{x}_s, \mathbf{x}', -t) * \nabla p(\mathbf{x}, \mathbf{x}', t) \right. \\ \left. - p(\mathbf{x}, \mathbf{x}', t) * \nabla p(\mathbf{x}_s, \mathbf{x}', -t) \right] \cdot \mathbf{n}(\mathbf{x}') \end{aligned} \quad (23)$$

Imaging

Rename $\mathbf{x}_s$ to $\mathbf{x}'$ and vice versa

$$\begin{aligned} p(\mathbf{x}', \mathbf{x}, -t) * h(t) - p(\mathbf{x}', \mathbf{x}, t) * h(-t) = \\ \int dS(\mathbf{x}_s) \left[p(\mathbf{x}', \mathbf{x}_s, -t) * \nabla p(\mathbf{x}, \mathbf{x}_s, t) \right. \\ \left. - p(\mathbf{x}, \mathbf{x}_s, t) * \nabla p(\mathbf{x}', \mathbf{x}_s, -t) \right] \cdot \mathbf{n}(\mathbf{x}') \end{aligned} \quad (24)$$

Interferometry



Imaging

Define $r(\mathbf{x}', \mathbf{x}, t) = p(\mathbf{x}', \mathbf{x}, -t) * h(t) - p(\mathbf{x}', \mathbf{x}, t) * h(-t)$

$$\begin{aligned} r(\mathbf{x}', \mathbf{x}, t) = \\ \int dS(\mathbf{x}_s) \int d\tau \left[p(\mathbf{x}', \mathbf{x}_s, t + \tau) \nabla p_0(\mathbf{x}, \mathbf{x}_s, \tau) \right. \\ \left. - p_0(\mathbf{x}, \mathbf{x}_s, t + \tau) \nabla p(\mathbf{x}', \mathbf{x}_s, \tau) \right] \cdot \mathbf{n}(\mathbf{x}') \end{aligned} \quad (25)$$

$$\begin{aligned} r(\mathbf{x}', \mathbf{x}, t = 0) = \\ \int dS(\mathbf{x}_s) \int d\tau \left[p(\mathbf{x}', \mathbf{x}_s, \tau) \nabla p_0(\mathbf{x}, \mathbf{x}_s, \tau) \right. \\ \left. - p_0(\mathbf{x}, \mathbf{x}_s, \tau) \nabla p(\mathbf{x}', \mathbf{x}_s, \tau) \right] \cdot \mathbf{n}(\mathbf{x}') \end{aligned} \quad (26)$$

p_0 : Forward wavefield

Imaging

The final imaging formula can be interpreted as

1. The left hand side is the wavefield at position $\mathbf{x}'$ due to a source at position $\mathbf{x}$. These two positions are arbitrary, so the sources (and receivers) are virtual.
2. The two terms on the right hand side is the cross-correlation between the extrapolated data and forward modelled data. Both the pressure and the derivative of the pressure are needed as data.

Imaging

Migration consists in:

1. Compute the forward wavefield $p(\mathbf{x}, \mathbf{x}_s, t)$ from equation (16)
2. Compute the backward wavefield $p(\mathbf{x}', \mathbf{x}_s, t)$ from equation (21).
3. Compute the image from equation (26)

Simplification

$$\begin{aligned} r(\mathbf{x}', \mathbf{x}, t = 0) = \\ \int dS(\mathbf{x}_s) \int d\tau \left[p(\mathbf{x}', \mathbf{x}_s, \tau) \nabla p_0(\mathbf{x}, \mathbf{x}_s, \tau) \right. \\ \left. - p_0(\mathbf{x}, \mathbf{x}_s, \tau) \nabla p(\mathbf{x}', \mathbf{x}_s, \tau) \right] \cdot \mathbf{n}(\mathbf{x}') \end{aligned} \quad (27)$$

Horizontal receiver implies

$$\begin{aligned} r(\mathbf{x}', \mathbf{x}, t = 0) = \\ \int dS(\mathbf{x}_s) \int d\tau \left[p(\mathbf{x}', \mathbf{x}_s, \tau) \partial_z p_0(\mathbf{x}, \mathbf{x}_s, \tau) \right. \\ \left. - p_0(\mathbf{x}, \mathbf{x}_s, \tau) \partial'_z p(\mathbf{x}', \mathbf{x}_s, \tau) \right] \end{aligned} \quad (28)$$

Simplification

$\partial_z p(\mathbf{x}', \mathbf{x}_s, t) = 0$ (No recorded pressure gradient)

$$r(\mathbf{x}', \mathbf{x}, t = 0) = \int dS(\mathbf{x}_s) \int d\tau p(\mathbf{x}', \mathbf{x}_s, \tau) \partial_z p_0(\mathbf{x}, \mathbf{x}_s, \tau) \quad (29)$$

Classical Imaging condition

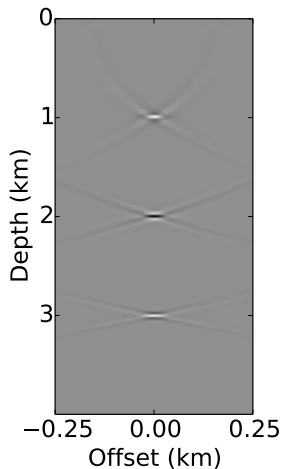
For $\mathbf{x}' = \mathbf{x}$ and by ignoring ∂_z this is the classical imaging condition (Claerbout, 1971)

$$r_c(\mathbf{x}) = \sum_{\mathbf{x}_s} \sum_{\tau} p_0(\mathbf{x}, \mathbf{x}_s, \tau) p(\mathbf{x}, \mathbf{x}_s, \tau)$$

Ignoring ∂_z implies an unfocused image with less than optimal resolution and incorrect amplitudes.

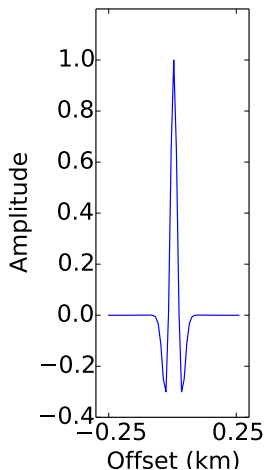
Numerical examples

Common image point gather (CIP) in the center of the model
Classical imaging condition:



Numerical examples

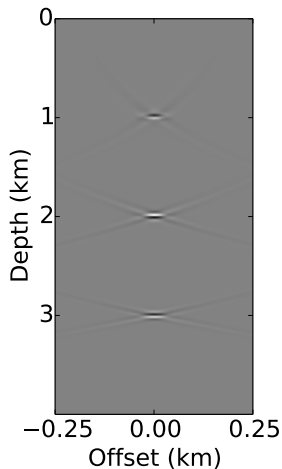
Horizontal profile through reflector at 1000m depth



Numerical examples

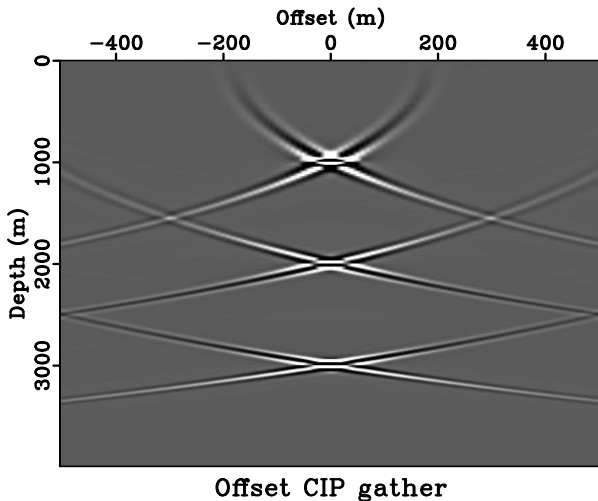
Common image point gather (CIP) in the center of the model.

New imaging condition:



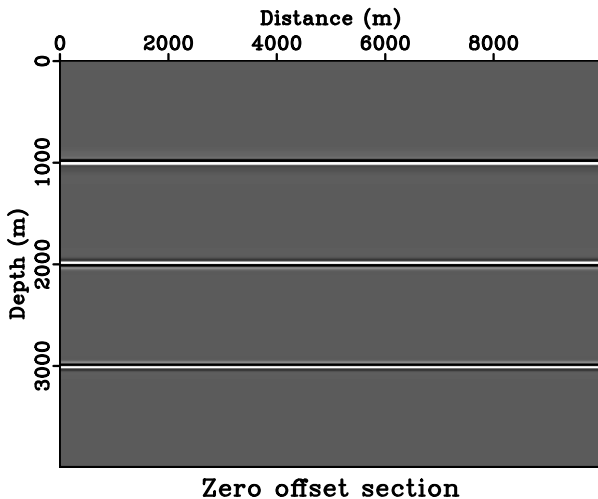
Numerical example

Reflectivity $p(\mathbf{x} - \mathbf{x}', t = 0)$ at all depths using new imaging condition



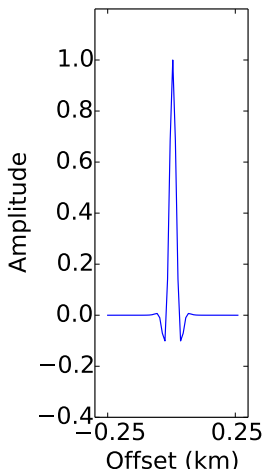
Numerical example

Full section $r(\mathbf{x} - \mathbf{x}' = 0, t)$ at all depths using new imaging condition



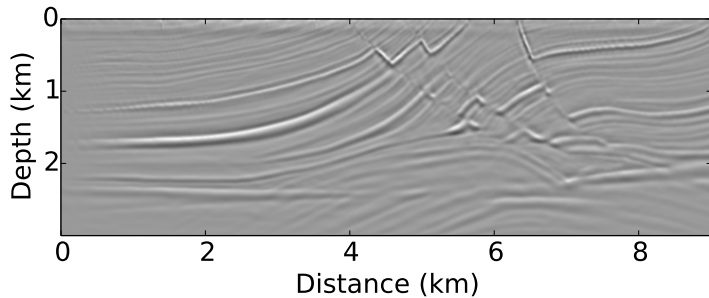
Numerical examples

Horizontal profile through reflector at 1000m depth



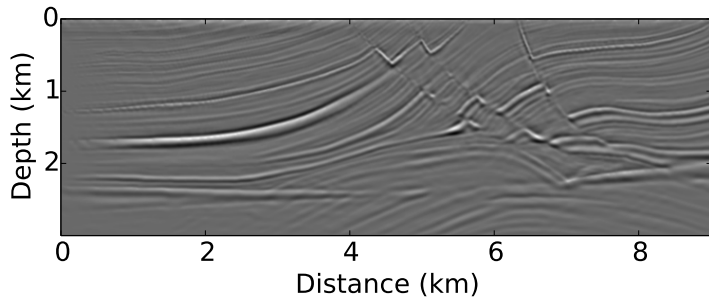
Numerical example

Conventional imaging condition:



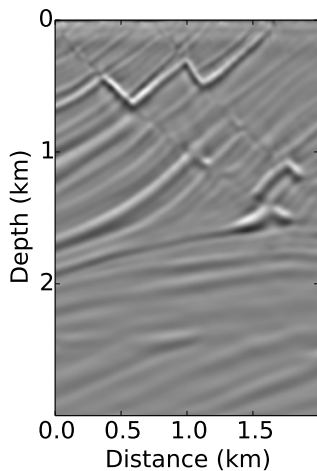
Numerical example

New imaging condition:



Numerical example

Conventional imaging condition:



Numerical example

New imaging condition:

